Week 1 –Project selection and initiation (Lecture 2)

COS30049-Computing Technology Innovation Project

Ricky Dong
rdong@swin.edu.au

Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.



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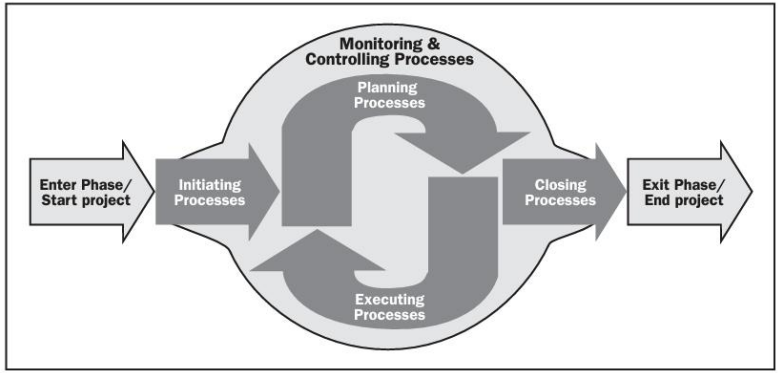
Objectives of Today

- Steps of Project Selection
- What is a Business Case
- Some Reasons for Choosing the Wrong Project
- Strategic Planning
- Web Design

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Recall of the Project Management Processes

1. ***Selection and Initiation***
2. Planning
3. Execution
4. Monitoring & Controlling
5. Closing
   - Closing iteration
   - Closing project



Project Management Process,

Project Management Institute, 2001, p.40

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Selection and initiation (A1)

Proper project selection is crucial for meeting the business's needs. Inadequate selection processes significantly reduce the chances of project success.

To increase the likelihood of project success, in the initial phase, we should：

- Analyse **Business Needs** and **Strategic Direction**
  - **Understand the organization's goals and objectives.**
  - Align the project with the strategic direction.
- Determine Project Nature and Scope
  - **Define the project's boundaries and deliverables.**
  - Identify key requirements and constraints.
- **Evaluate Project Value to the Business**
  - Assess the financial implications and potential returns.
  - Develop a **business case**. **(Why you need to do this? What is the value?)**

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Establishing an Overall Picture

To establish a comprehensive understanding of the project, consider the following key areas:

- **Evaluating Business Value (A1)**
  - **Alignment with Strategic Plans:**
    - How does the project align with the long-term strategic goals of the business?
  - **Potential Benefits:**
    - Will the project introduce a new product?
    - Can it improve customer support?
    - Will it enhance productivity?
    - Is it likely to reduce operational costs?
- **Defining Project Nature and Scope (A1)**
  - **Project scope:**
    - The total amount of work, goals, tasks, and deliverables needed to complete a project
  - **Cost Justification:**
    - Are the costs of the project, including development and support, justified by the expected benefits?
  - **Measurable Benefits:**
    - What specific, measurable advantages does the project offer compared to its expenditures?
- **Conducting Risk Assessment (A1)**
  - **Risk Identification:**
    - Is the project associated with high risks?
  - **Mitigation Strategies:**
    - Identify potential risks and develop strategies to mitigate them.
  - **Anticipated Challenges:**
    - What challenges or issues could arise during the project?

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What is a Business Case (A1)

- **Purpose**: Helps decision-makers determine if a project is worth pursuing and secure approval and funding.
- **Value**: Ensures the project aligns with strategic goals and provides a foundation for planning and execution.
- **Content**: Includes an executive summary, problem statement, analysis of options, cost-benefit analysis, risk assessment, implementation plan, and conclusion with recommendations.
- **Target Audience**: Executives, senior management, project sponsors, stakeholders, and project managers.

Business Case

Project Name	Project Manager
Client	**Duration**

Executive Summary	• Write a short version of each of the following sections in your business case.
Mission Statement	• Define the vision, goals and objectives of the project.
Product/ Service	• Explain what the product or service and how it fits a niche or serves a need.
Project Definition	• Provide general information about the project, such as a project plan outline.
Project Organization	• What is the structure of the project, such as functional, matrix, projectized or composite

Reasons for choosing the wrong projects?

Several major reasons why organizations choose the wrong projects include:

- **Bias and Errors in Judgment**
  - Subjective preferences and personal biases affecting decisions.
- **Ineffective Project Portfolio Management**
  - Failure to establish a robust framework for managing project portfolios.
- **Lack of Proper Metrics**
  - Absence of accurate metrics for evaluating project value.
- **Inadequate Risk Assessment**
  - Inability to properly assess and value project risks.
- **Misidentification of Efficient Project Portfolios**
  - Failure to identify and select projects that align with the "efficient frontier."

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Strategic Planning/Alignment (A1)

A strategic plan is a document used by company leaders to outline the company's long-term vision, goals, and objectives, typically focusing on a timeframe of 3-5 years.

The development of strategies (projects) must focus on what is needed by the organisation to meet the strategic plan’s goals and objectives

Ask Yourself:

Does the proposed project deliver a product or service that aligns with the objectives defined in the strategic plan?



Strategic Planning: SWOT analysis (A1)

- A commonly used tool to build a strategic plan is the SWOT analysis. It is an information-gathering technique that evaluates external influences against internal capabilities.
- Strengths: Internal attributes that support achieving objectives.
- Weaknesses: Internal attributes that hinder achieving objectives.
- Opportunities: External factors that the organization can capitalize on.
- Threats: External factors that could jeopardize the organization's success.

Related video on how to perform a SWOT Analysis:
https://www.youtube.com/watch?v=I_6AVRGLXGA

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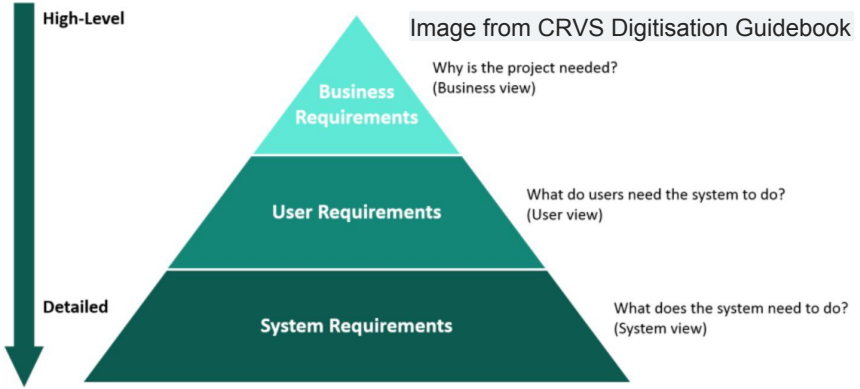
Step 1 of Web Design: How to define requirements ? (A1)

The following processing ideas can be used when defining requirements and producing a project proposal

In your assignment 1, you need to list functional and non-funcitonal requirements

- Functional requirements: What the system must do. Features, specific behaviors, and tasks the system must execute (e.g., users can log in via email).
- Non-functional requirements: How the system must perform. Quality attributes like speed, security, reliability, and usability (e.g., pages load in under 2 seconds).

Define the context and objectives of the project: Understand the origin and objectives of the project, and identify why the project was initiated and what it is expected to achieve.

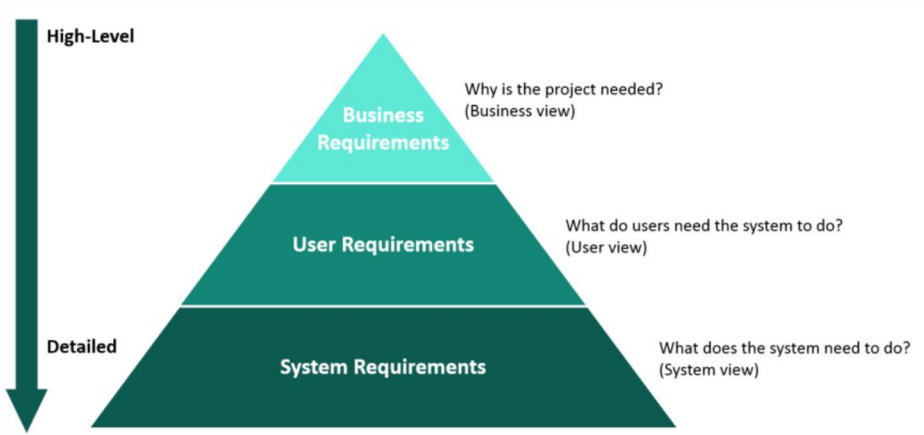


Requirements Hierachy

Step 1 of Web Design: How to define requirements ? (A1)

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Gather stakeholder requirements: Communicate with various stakeholders to understand their needs and expectations, and ensure that the project fully considers the requirements of all key stakeholders.



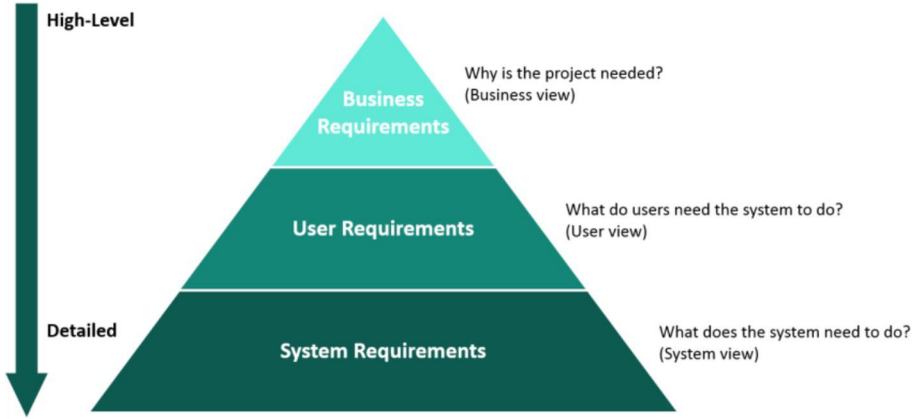
Requirements Hierachy

Image from CRVS Digitisation Guidebook

Step 1 of Web Design: How to define requirements? (A1)

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Analyse the scope of the project: Define the boundaries of the project, including the functions to be covered, the target users, the geographic scope, etc.



Requirements Hierachy

Image from CRVS Digitisation Guidebook

Step 2 of Web Design: Why Prototype Design ? (A1)

Prototyping plays a crucial role in projects for several reasons:

- Clarify Requirements: Prototyping helps teams and stakeholders better understand project requirements. Through visualisation, prototyping allows everyone to have a clearer understanding of the project's functionality and interfaces, avoiding confusion and misunderstanding in the understanding of requirements.



Image from uxpin.com

Step 2 of Web Design: Why Prototype Design? (A1)

Prototyping plays a crucial role in projects for several reasons:

- Identify problems in advance: Through prototyping, project teams can identify and solve potential problems before development. This helps to reduce the cost of modifications and adjustments at a later stage of the project, while increasing the likelihood of project success.



Image from uxpin.com

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Step 2 of Web Design: Why Prototype Design? (A1)

Prototyping plays a crucial role in projects for several reasons:

- Ensure consistency: prototyping ensures that the entire team is aligned on the goals and style of the project. This is especially important in large projects or when multiple people are working together to avoid disagreements between multiple team members.

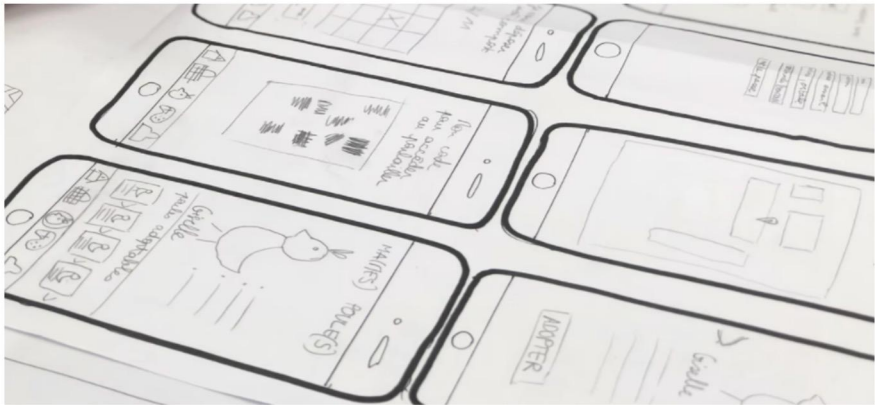


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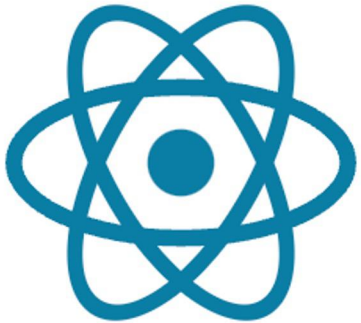
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Step 3 of Web Design: Building Your Web with React.js (A3)

React.js is a popular JavaScript library developed and maintained by Facebook that has many features that make it ideal for building modern front-end applications. Here are the key features of React.js:

- Componentized development: React.js uses a componentized development model to split the application into multiple independent, reusable components. Each component has its own state and behaviour and can be nested and combined.

- Virtual DOM: React.js introduces the concept of a virtual DOM, which is a lightweight copy of the actual DOM similar to the browser's. When data changes, React improves rendering performance and efficiency by comparing the differences between the virtual DOM and the actual DOM and minimally updating the actual DOM.



Thank You